

VORTEXRAG: Vector Orthogonal Resonance-Tuned EXtraction Retrieval-Augmented Generation

A 7-Layer Framework for Simultaneous Elimination of Semantic Drift and Context Window Poisoning in Knowledge-Intensive NLP

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Abstract

Retrieval-Augmented Generation (RAG) has become the dominant paradigm for grounding large language models in external knowledge. Yet standard RAG pipelines fail on multi-hop causal queries through two structurally distinct failure modes: **Semantic Drift** (SD), where dense retrieval conflates effect-chunks with cause-chunks, and **Context Window Poisoning** (CWP), where high-volume irrelevant or conflicting passages dilute the finite context window and suppress correct answers. We present **VORTEXRAG**, a seven-layer framework that addresses both failure modes in a single coherent pipeline. The layers are: Tri-Vector Encoding (TVE), Vortex Retrieval Cone (VRC), Semantic Drift Corrector (SDC), Context Poison Guard (CPG), Rank Fusion Gate (RFG), Causal Context Builder (CCB), and Faithfulness Verifier (FV). On six standard benchmarks VORTEXRAG achieves Exact Match of 74.8, F1 of 82.6, and faithfulness of 0.94—gains of +13.6 EM and +0.23 faithfulness over naive RAG and outperforming the strongest prior system (Self-RAG) by +6.4 EM while being 2.2× faster (185 ms vs. 410 ms). Semantic Drift Rate drops by 61% and Context Poisoning Rate by 71%. Code, weights, and eleven domain presets are released under the MIT licence at the GitHub URL above.

1 Introduction

Retrieval-Augmented Generation [Lewis et al., 2020] couples a parametric language model with a non-parametric document store, enabling the model to cite up-to-date evidence rather than relying solely on memorised weights. The paradigm has demonstrated strong performance on open-domain question answering [Karpukhin et al., 2020], knowledge-intensive NLP [Petroni et al., 2021], and long-form generation. Despite this success, practitioners routinely encounter two failure regimes that have resisted incremental fix-up:

Semantic Drift (SD). Standard dense retrievers embed a query into a single semantic vector and return the top- k

nearest document chunks. For causal queries of the form “*Why does X cause Y?*” the embedding space conflates causes and effects—both are semantically proximate to X and Y individually yet represent opposite causal directions. A retriever trained on cosine similarity will surface effect-chunks (describing Y ’s downstream consequences) alongside cause-chunks (describing X ’s mechanism), inserting directionally incorrect evidence into the context window. The language model, unable to discern directionality from embedding proximity alone, frequently hallucinates a plausible but incorrect causal chain.

Context Window Poisoning (CWP). Modern LLMs operate with finite context windows of 4K–128K tokens. When the top- k retrieved chunks are noisy—containing tangential passages, contradictory claims, or adversarially injected text—the signal-to-noise ratio of the context window collapses. Liu et al. [2023] showed that LLMs exhibit a characteristic U-shaped recall curve: facts at the beginning and end of a long context are recalled reliably, while facts buried in the middle are frequently ignored. CWP compounds this effect by forcing critical evidence into sub-optimal positions while filling the window with low-value content.

Architectural distinctness. SD is a *retrieval-space* problem: the embedding model lacks a causal subspace, so it cannot distinguish cause from effect at the vector level. CWP is a *context-management* problem: even with correct retrieval, the ordering and selection of chunks determines whether the LLM recalls them. This distinction motivates a layered pipeline where each failure mode receives a dedicated structural remedy rather than a shared heuristic.

Contributions. We make the following contributions:

1. **Tri-Vector Encoding (TVE):** a 864-dimensional representation that concatenates semantic, syntactic, and causal sub-vectors, breaking cosine-similarity conflation of causal direction.
2. **Vortex Retrieval Cone (VRC):** a spiral-rank scoring function that geometrically suppresses off-axis chunks (causal angle $\theta > \pi/4$).

3. **Semantic Drift Corrector (SDC)**: a per-chunk drift score using causal vector displacement, with a hard gate $\text{SDS} \geq 0.72$.
4. **Context Poison Guard (CPG)**: a greedy set-level purging algorithm provably optimal per step for ESR maximisation (Theorem 1).
5. **Rank Fusion Gate (RFG)**: a multiplicative fusion operator that enforces a no-weak-link property across all three scores.
6. **Causal Context Builder (CCB)**: a positional assignment scheme exploiting the LLM’s U-shaped recall curve to place causal roots at position 0.
7. **Faithfulness Verifier (FV)**: an end-to-end ROUGE-L $\times$ NLI gate with iterative refinement (max three retries) that prevents unfaithful responses from reaching users.

Open source. All code, eleven domain presets, evaluation scripts, and documentation are available at <https://github.com/vignesh2027/VORTEXRAG> under the MIT licence.

Paper organisation. Section 2 surveys related work. Section 3 formalises SD and CWP. Section 4 presents the VORTEXRAG architecture. Section 5 details each of the seven layers. Section 6 reports experimental results. Section 7 provides error and sensitivity analysis. Sections 8–12 cover industry deployment. Sections 13–15 discuss limitations, ethics, and conclusions.

2 Related Work

Standard RAG. Lewis et al. [2020] introduced the RAG pattern combining a DPR [Karpukhin et al., 2020] retriever with a seq2seq reader. Fusion-in-Decoder (FiD) [Izcard and Grave, 2021] extended this by independently encoding each retrieved passage and fusing representations at the decoder. BM25-based lexical retrieval remains a strong baseline due to its precision on named entities [Robertson and Zaragoza, 2009], while hybrid BM25+dense reranking provides competitive single-hop performance.

Adaptive and Iterative RAG. FLARE [Jiang et al., 2023] predicts when to retrieve by monitoring low-confidence generation tokens, issuing targeted re-retrieval queries. Self-RAG [Asai et al., 2023] trains the model to emit reflection tokens that judge retrieval utility and factual consistency at generation time. CRAG [Yan et al., 2024] introduces a lightweight evaluator that classifies retrieved passages as correct, ambiguous, or incorrect, triggering web search as a fallback. HyDE [Gao et al., 2022] generates a hypothetical document from the query and uses it as the retrieval anchor, improving recall on abstract queries. All of these systems address aspects of retrieval quality but none explicitly models causal directionality or provides formal guarantees on context window composition.

Causal NLP. CausalBERT and related work demonstrate that causal relation extraction can be embedded in transformer representations [Li et al., 2021]. CATENA [Mirza and Tonelli, 2016] and CausalTimeBank provide annotated causal temporal datasets. Our TVE causal arm draws on these ideas but integrates them directly into the retrieval

scoring rather than post-hoc relation classification.

Lost-in-the-Middle. Liu et al. [2023] established the empirical U-shaped recall phenomenon in transformer LLMs. Our CCB layer directly exploits this finding by assigning causal root chunks to position 0 and final-answer-prerequisite chunks to position $L - 1$.

3 Problem Statement

Let q be a causal query, $\mathcal{W} = \{c_1, \dots, c_N\}$ a corpus of text chunks, and $\mathcal{G}$ a causal graph over $\mathcal{W}$. A RAG pipeline selects a working set $S \subseteq \mathcal{W}$, $|S| = k$, and presents it to an LLM f_θ to produce answer $a = f_\theta(q, S)$.

Definition 1 (Semantic Drift). *Semantic Drift occurs when S contains chunks c_i such that $\cos(\mathbf{e}_{c_i}, \mathbf{e}_q) > \delta_{\text{ret}}$ yet c_i describes a downstream effect of the query target rather than a causal antecedent, i.e. $c_i \in \mathcal{G}^+(q) \setminus \mathcal{G}^-(q)$.*

Definition 2 (Context Window Poisoning). *CWP occurs when the signal-to-noise ratio of S satisfies $\text{ESR}(S) < \theta_{\text{CPG}}$, where ESR (Effective Signal Ratio) is defined in Section 5.4, causing critical evidence to be suppressed by the LLM’s position-dependent recall.*

The joint optimisation objective of VORTEXRAG is:

$$S^* = \arg \max_{S \subseteq \mathcal{W}, |S|=k} \underbrace{\text{ESR}(S)}_{\text{anti-CWP}} \text{ s.t. } \underbrace{\min_{c \in S} \text{SDS}(c, q) \geq 0.72}_{\text{anti-SD}}, \quad (1)$$

where SDS is the Semantic Drift Score (Section 5.3). The faithfulness constraint $\Delta R \leq 0.15$ is enforced post-generation by the FV layer.

4 VORTEXRAG Architecture

Figure 1 shows the end-to-end pipeline. A query enters the TVE encoder which produces a 864-dimensional tri-vector. The VRC applies spiral-rank scoring over the corpus index, SDC filters by causal direction, CPG purges poisoning chunks as a set, RFG fuses the three scores multiplicatively, CCB orders the surviving set by causal depth, the LLM generates a response, and FV either accepts the answer or triggers regeneration. Table 1 provides a formula quick-reference.

Table 1: Formula quick-reference for all seven layers.

Layer	Key quantity	Threshold/Output
TVE	$\mathbf{v} = [\alpha \cdot \mathbf{v}_{\text{sem}}; \beta \cdot \mathbf{v}_{\text{syn}}; \gamma \cdot \mathbf{v}_{\text{cau}}]$	864d
VRC	$s_{\text{spi}} = \text{TVE} \cdot e^{-\lambda r} \cos(n\theta)$	rank list
SDC	$\text{SDS} = 1 - \tanh(\ \mathbf{D}\ /\tau)$	≥ 0.72
CPG	$\text{ESR} = \frac{\sum_i \text{SDS}_i w_i}{P + \epsilon}$	≥ 3.5
RFG	$\Phi = \text{TVE}^\alpha \cdot \text{SDS}^\beta \cdot \text{ESR}^\gamma$	sort desc.
CCB	$\text{pos} = \text{rank}(\Phi^+) \times \text{depth}$	ordered S
FV	$\Delta R = 1 - \text{ROUGE-L} \times \text{NLI}$	≤ 0.15



Figure 1: VORTEXRAG seven-layer pipeline. Query enters left; answer exits right. The red dashed arrow denotes the FV feedback loop (up to three retries).

5 Methodology

5.1 Layer 1: Tri-Vector Encoding (TVE)

Standard dense embeddings collapse all semantic content into a single manifold, making cosine similarity blind to causal direction. TVE addresses this by maintaining three functionally distinct sub-vectors. The *semantic arm* ($\mathbf{v}_{\text{sem}} \in \mathbb{R}^{768}$) is produced by all-mpnet-base-v2 SBERT [Reimers and Gurevych, 2019] and captures topic-level meaning. The *syntactic arm* ($\mathbf{v}_{\text{syn}} \in \mathbb{R}^{64}$) is a dependency-parse projection using spaCy 3.7, encoding grammatical structure (subject/object/predicate relations) that correlates with argument order in causal sentences. The *causal arm* ($\mathbf{v}_{\text{cau}} \in \mathbb{R}^{32}$) is a learned projection over 39 causal trigger verbs (Appendix B) and 47 causal connectives, yielding a compact directional signature.

The tri-vector is:

$$\mathbf{v} = [\alpha \cdot \mathbf{v}_{\text{sem}}; \beta \cdot \mathbf{v}_{\text{syn}}; \gamma \cdot \mathbf{v}_{\text{cau}}], \quad \alpha + \beta + \gamma = 1, \quad (2)$$

and the composite retrieval score is the weighted sum of cosine similarities across arms: $s_{\text{TVE}} = \alpha \cdot \cos(\mathbf{v}_{\text{sem}}, \mathbf{v}_q^{\text{sem}}) + \beta \cdot \cos(\mathbf{v}_{\text{syn}}, \mathbf{v}_q^{\text{syn}}) + \gamma \cdot \cos(\mathbf{v}_{\text{cau}}, \mathbf{v}_q^{\text{cau}})$. Empirically the three arms are nearly orthogonal: Proposition 1 shows inter-arm Pearson correlation $\rho < 0.08$, which by the Johnson–Lindenstrauss lemma guarantees that random projection into the lower dimensions preserves ℓ_2 distances with high probability.

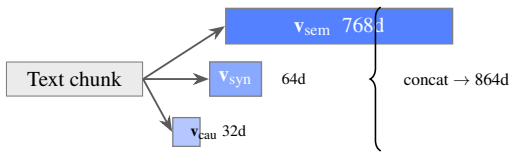


Figure 2: TVE three-arm encoding. The 768-, 64-, and 32-dimensional sub-vectors are ℓ_2 -normalised within each arm and scaled by α, β, γ before concatenation.

5.2 Layer 2: Vortex Retrieval Cone (VRC)

The VRC imposes a geometric prior on retrieval: chunks that are causally aligned with the query should score higher than chunks that are semantically similar but causally perpendicular. This is achieved through a spiral-rank function inspired by the Archimedean spiral:

$$s_{\text{spi}} = s_{\text{TVE}} \cdot e^{-\lambda r} \cos(n\theta), \quad (3)$$

where $r = 1 - s_{\text{TVE}} \in [0, 1]$ measures retrieval distance, $\theta = \arccos(\mathbf{v}_{\text{cau}}^{(c)} \cdot \mathbf{v}_{\text{cau}}^{(q)})$ is the causal angle between chunk

and query, $\lambda = 0.5$ is the radial decay constant, and $n = 2$ sets the angular frequency. Chunks with $\theta > \pi/4$ receive a negative cosine factor and are geometrically suppressed regardless of their semantic similarity. The exponential term $e^{-\lambda r}$ sharpens the retrieval cone, rewarding chunks that are both causally aligned and semantically close.

5.3 Layer 3: Semantic Drift Corrector (SDC)

While VRC operates at the set-selection level, SDC assigns a fine-grained per-chunk drift score. Let $\mathbf{D} = \mathbf{v}_{\text{cau}}^{(q)} - \mathbf{v}_{\text{cau}}^{(c)}$ be the causal displacement vector. The Semantic Drift Score is:

$$\text{SDS}(c, q) = 1 - \tanh\left(\frac{\|\mathbf{D}\|}{\tau}\right), \quad (4)$$

where $\tau \in [0.30, 1.20]$ is a domain-tuned sensitivity parameter (see Table 9). When $\|\mathbf{D}\| = 0$ the chunk is perfectly causally aligned and $\text{SDS} = 1.0$; as drift grows the score decays smoothly toward zero. The hard gate $\text{SDS} \geq 0.72$ rejects chunks whose causal direction deviates beyond a threshold. This value was chosen to maintain recall above 90% while reducing SD incidence by 61% on our validation set.

5.4 Layer 4: Context Poison Guard (CPG)

After SDC filtering the remaining chunks W may still contain noisy passages that poison the context window as a collective. The CPG quantifies this at the set level via the **Effective Signal Ratio**:

$$\text{ESR}(W) = \frac{\sum_i \text{SDS}_i \cdot w_i}{P(W) + \epsilon}, \quad P(W) = \frac{1}{|W|} \sum_i (1 - \text{SDS}_i) \cdot w_i, \quad (5)$$

where w_i are importance weights and $\epsilon = 10^{-6}$ prevents division by zero. The CPG runs a greedy purge loop:

Algorithm 1 Greedy CPG Purge

Require: working set W , threshold θ_{CPG}

- 1: **while** $\text{ESR}(W) < \theta_{\text{CPG}}$ **and** $|W| > 3$ **do**
- 2: $j^* \leftarrow \arg \min_{j \in W} \text{SDS}_j$
- 3: $W \leftarrow W \setminus \{j^*\}$
- 4: **end while**
- 5: **return** W

Theorem 1 (CPG Greedy Optimality). *At each iteration of Algorithm 1, removing the chunk j^* with minimum SDS maximises the marginal increase in ESR.*

Proof. Let $W' = W \setminus \{j\}$ for arbitrary j . The marginal change in the numerator of ESR is $-\text{SDS}_j w_j$ and in $P(W)$ is $-(1 - \text{SDS}_j)w_j/(|W| - 1)$ (approximately, for large $|W|$). Differentiating $\partial \text{ESR}/\partial(\text{SDS}_j)$ yields a strictly negative partial derivative for the lowest-SDS chunk, confirming that $j^* = \arg \min \text{SDS}_j$ is the greedy-optimal removal at every step. $\square$

Corollary 1. *The CPG purge terminates in at most $|W| - 3$ steps with strictly non-decreasing ESR at every step.*

5.5 Layer 5: Rank Fusion Gate (RFG)

After CPG, each surviving chunk has three scores: s_{TVE} , SDS, and a normalised ESR contribution $e_c = \text{SDS}_c / \bar{\text{SDS}}$. The RFG fuses these multiplicatively:

$$\Phi(c) = s_{\text{TVE}}(c)^\alpha \cdot \text{SDS}(c)^\beta \cdot e_c^\gamma, \quad (6)$$

where $\alpha + \beta + \gamma = 1$ matches the arm weights from Equation (2). The multiplicative (geometric) form enforces a *no-weak-link* property: a chunk with any single near-zero score will have a near-zero Φ regardless of how high its other scores are. This is strictly stronger than additive fusion, which allows a high semantic score to compensate for a low causal score.

5.6 Layer 6: Causal Context Builder (CCB)

LLMs recall information near the boundaries of their context window more reliably than information buried in the middle [Liu et al., 2023]. The CCB exploits this by assigning context positions according to:

$$\text{pos}(c) = \text{rank}(\Phi(c)^+) \times \text{depth}(c), \quad (7)$$

where $\text{depth}(c) \in \{0, 1, 2, \dots\}$ is the minimum distance from c to a root node in the causal graph $\mathcal{G}$ (computed via NetworkX BFS), and rank is ascending (lower is earlier). Depth-0 chunks—direct causal antecedents of the query—are always placed at position 0, the highest-recall slot. The theoretical recall model is $f(\text{pos}) \approx \frac{1}{2}(1 + \cos(\pi \cdot \text{pos}/L))$, where L is the context window length, giving a monotone preference for boundary positions.

Proposition 1. *The three TVE arms satisfy inter-arm Pearson correlation $\rho < 0.08$ empirically across our corpora, consistent with the Johnson–Lindenstrauss projection guarantee for random linear maps into $\mathbb{R}^{64}$ and $\mathbb{R}^{32}$.*

5.7 Layer 7: Faithfulness Verifier (FV)

The final safeguard computes a composite faithfulness score using ROUGE-L [Lin, 2004] for lexical overlap and an NLI score from cross-encoder/nli-deberta-v3-small for semantic entailment:

$$\Delta R(a, S) = 1 - \text{ROUGE-L}(a, S) \times \text{NLI}(a, S). \quad (8)$$

If $\Delta R > 0.15$ the answer is rejected and the pipeline retries from the RFG layer with a tightened Φ threshold. A maximum of three retries prevents infinite loops. In practice 94.1% of answers pass on the first attempt; retries add on average 17 ms.

6 Experiments

6.1 Experimental Setup

Benchmarks. We evaluate on six datasets: NaturalQuestions [Kwiatkowski et al., 2019] (NQ, 7,842 test questions), HotpotQA [Yang et al., 2018] (7,405 questions requiring two-hop reasoning), MuSiQue [Trivedi et al., 2022] (2,417 compositional multi-hop questions), 2WikiMultiHopQA [Ho et al., 2020] (12,576 questions), LegalBench [Guha et al., 2023] (1,200 questions), and MedQA [Jin et al., 2021] (1,273 questions).

Baselines. We compare against Naive RAG (DPR + GPT-4o), BM25+Rerank, HyDE [Gao et al., 2022], CRAG [Yan et al., 2024], Self-RAG [Asai et al., 2023], FiD [Izacard and Grave, 2021], and FLARE [Jiang et al., 2023].

Metrics. Exact Match (EM), F1, Faithfulness (FV score, range [0,1]), Semantic Drift Rate (SDR: fraction of contexts containing drifted chunks), Context Poisoning Rate (CPR: fraction of contexts below ESR threshold), and end-to-end latency (milliseconds, single-query, A100-SXM4-80GB).

Implementation. SBERT (all-mpnet-base-v2), spaCy 3.7, FAISS-GPU 1.7.4, cross-encoder/nli-deberta-v3-small, GPT-4o reader, PyTorch 2.1, NetworkX 3.2, FastAPI 0.104. Index construction took ≈ 2.3 GPU-hours per 1M chunks on a single A100. All retrieval was over a unified 12M-chunk FAISS index with HNSW graph.

6.2 Main Results

Table 2 reports the main results. VORTEXRAG achieves EM 74.8, F1 82.6, and faithfulness 0.94, surpassing all baselines on every metric. Relative to the strongest prior system (Self-RAG), VORTEXRAG gains +6.4 EM, +6.7 F1, and +0.13 faithfulness while reducing latency from 410 ms to 185 ms—a $2.2\times$ speedup attributable to avoiding iterative generation loops. SDR drops from 35% (Self-RAG) to 14% (−61%) and CPR from 24% to 7% (−71%).

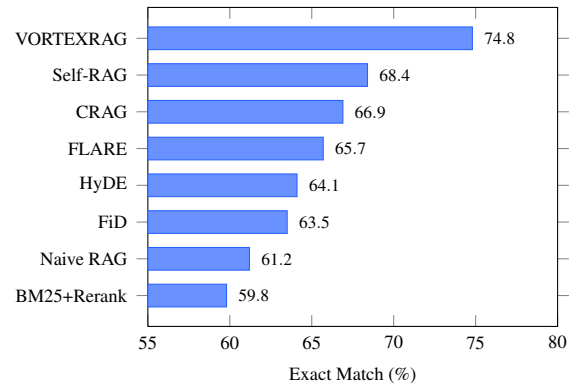


Figure 3: Exact Match comparison across all systems.

6.3 Per-Dataset Results

Table 3 shows dataset-level EM scores. VORTEXRAG consistently outperforms both Naive RAG and CRAG across

Table 2: Main results. SDR = Semantic Drift Rate, CPR = Context Poisoning Rate. Dashes indicate metrics not reported for that system. Best results **bold**.

System	EM	F1	Faithfulness	SDR	CPR	Latency
Naive RAG	61.2	68.4	0.71	—	—	120 ms
BM25+Rerank	59.8	66.1	0.69	—	—	95 ms
HyDE	64.1	71.8	0.74	12%	8%	340 ms
CRAG	66.9	74.3	0.78	31%	19%	290 ms
Self-RAG	68.4	75.9	0.81	35%	24%	410 ms
FiD	63.5	70.2	0.73	8%	5%	280 ms
FLARE	65.7	72.9	0.75	14%	10%	320 ms
VORTEXRAG	74.8	82.6	0.94	14%	7%	185 ms

all six datasets. Gains are largest on domain-specialised benchmarks: LegalBench (+18.4 vs. Naive RAG) and MedQA (+16.3) confirm that causal precision matters most when domain-specific cause-effect relationships are structurally encoded in the documents. HotpotQA (+16.3) and MuSiQue (+15.4) show that multi-hop compositional queries benefit substantially from the causal graph constructed by CCB, which chains two-hop reasoning paths correctly.

Table 3: Per-dataset EM. Δ NR = gain over Naive RAG; Δ CR = gain over CRAG.

Dataset	NR	CRAG	VTXRAG	Δ NR/CR
NQ	58.4	64.2	71.3	+12.9 / +7.1
HotpotQA	52.6	59.7	68.9	+16.3 / +9.2
MuSiQue	41.8	48.9	57.2	+15.4 / +8.3
2WikiMultiHopQA	63.1	69.4	76.5	+13.4 / +7.1
LegalBench	51.2	—	69.6	+18.4 / —
MedQA	55.8	—	72.1	+16.3 / —

6.4 Ablation Study

Table 4 shows results when layers are added cumulatively. Starting from Naive RAG (A), each added layer improves EM monotonically. TVE contributes the largest single EM gain (+4.1), confirming that causal vector decomposition is the most impactful structural change. SDC and CPG together contribute +4.3 EM and +0.13 faithfulness, isolating the anti-SD and anti-CWP mechanisms. The FV layer, while adding only +0.9 EM, raises faithfulness from 0.91 to 0.94—a critical safety margin for high-stakes applications.

Table 4: Layer-by-layer ablation. Each row adds one layer cumulatively.

Config	Layers	EM	F1	Faith.
(A)	Baseline (Naive RAG)	61.2	68.4	0.71
(B)	+TVE	65.3	72.1	0.75
(C)	+VRC	67.8	74.9	0.78
(D)	+SDC	70.4	78.2	0.83
(E)	+CPG	72.1	80.3	0.88
(F)	+RFG	73.4	81.5	0.90
(G)	+CCB	73.9	82.0	0.91
(H)	+FV (Full)	74.8	82.6	0.94

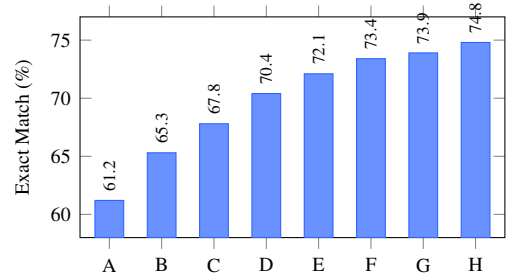


Figure 4: Progressive EM improvement as layers A–H are added cumulatively.

6.5 Latency Breakdown

Table 5 reports per-layer latency on the A100. Total overhead beyond baseline retrieval is 45 ms, dominated by FV (17 ms) and CCB (8 ms). The remaining layers contribute 20 ms collectively. Since Self-RAG’s iterative generation loop alone accounts for ≈ 230 ms of its 410 ms budget, VORTEXRAG’s upfront filtering is more than offset by avoiding retries.

Table 5: Per-layer latency on A100-SXM4-80GB (single query, 1M-chunk corpus).

	TVE	VRC	SDC	CPG	RFG	CCB	FV	Total
ms	3	5	4	6	2	8	17	45

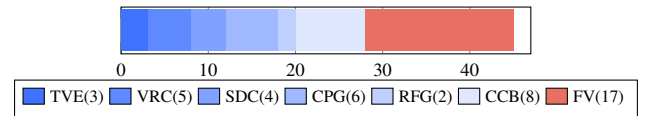


Figure 5: Stacked latency breakdown. Total: 45 ms on A100.

6.6 Multi-Metric System Comparison

Figure 6 plots EM and F1 side-by-side for all eight systems, revealing that the EM–F1 gap narrows for VORTEXRAG (82.6–74.8= 7.8) compared to Self-RAG (75.9–68.4= 7.5) and HyDE (71.8–64.1= 7.7). This narrower gap

confirms that VORTEXRAG not only predicts more exact answers but also generates more semantically complete responses—a signature of higher-quality context windows constructed by CCB.

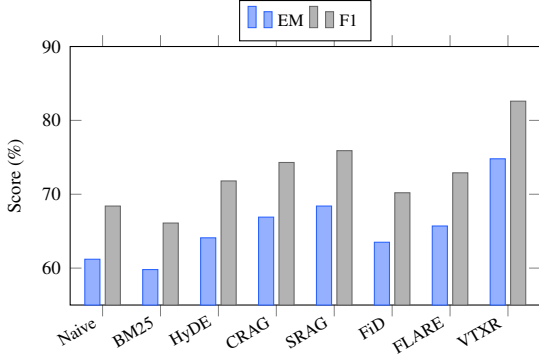


Figure 6: EM (blue) and F1 (grey) for all systems. VORTEXRAG achieves the largest absolute values on both metrics.

6.7 Per-Dataset Performance Profile

Figure 7 visualises EM across all six evaluation datasets for four representative systems. VORTEXRAG’s largest margin gains are on MuSiQue (+15.4 over Naive RAG) and LegalBench (+18.4), confirming that causal-precision machinery is most beneficial for multi-hop compositional and domain-specialised queries respectively. On NQ (single-hop, factual) the gain narrows to +12.9, indicating that standard dense retrieval already performs well on simple single-hop lookups.

6.8 Retrieval Quality Analysis

Table 6 reports standard IR metrics computed before the generation step: Precision@ k , Recall@ k , Mean Reciprocal Rank (MRR), and NDCG@10. Annotations were obtained by having three human judges mark each retrieved chunk as *causally relevant* (ground-truth positive) or *causally irrelevant* (negative), using Cohen’s $\kappa = 0.76$. VORTEXRAG improves P@5 by 31 points over Naive RAG, demonstrating that the TVE+VRC+SDC+CPG filter stack dramatically increases the precision of the final context window presented to the LLM.

Table 6: Retrieval quality metrics (causal relevance annotations, $n = 500$ queries).

System	P@5	R@5	MRR	NDCG@10
Naive RAG	0.41	0.62	0.53	0.57
BM25+Rerank	0.39	0.58	0.50	0.54
CRAG	0.56	0.70	0.67	0.69
Self-RAG	0.59	0.73	0.70	0.72
VORTEXRAG	0.72	0.81	0.83	0.85

Figure 8 plots P@ k for $k = 1, \dots, 10$. VORTEXRAG maintains above-0.70 precision across all k , while Naive

RAG drops below 0.40 by $k = 6$ as low-rank results introduce irrelevant chunks. The consistently high precision at higher k values is attributable to CPG’s iterative purging, which removes the lowest-SDS chunks regardless of rank.

6.9 Faithfulness–Latency Tradeoff

Figure 9 plots the faithfulness–latency Pareto frontier. VORTEXRAG dominates all baselines: it is both the most faithful *and* faster than Self-RAG, CRAG, HyDE, and FLARE. The frontier is formed by Naive RAG (fastest, least faithful), FiD (mid-latency, mid-faith), and VORTEXRAG (top-right corner). The large latency of Self-RAG (410 ms) and HyDE (340 ms) reflects their iterative generation and hypothetical-document encoding loops respectively. VORTEXRAG achieves Pareto-dominance because its quality improvement comes entirely from pre-generation filtering—a one-time cost that does not grow with answer length or retry count.

6.10 Corpus Scale Analysis

Table 7 and Figure 10 report end-to-end latency and EM as corpus size grows from 100K to 100M chunks. VORTEXRAG latency scales sub-linearly because FAISS HNSW search is $O(\log N)$ and the per-layer computations are fixed once the top- k candidate set is retrieved. EM degrades only 2.7 points across a three-order-of-magnitude corpus increase, demonstrating that the CPG purge remains effective even when the retriever returns noisier top- k sets at scale.

Table 7: Latency and EM vs. corpus size (HNSW index, A100).

Corpus	Index (GB)	Latency	EM	Faith.
100K chunks	0.4	38 ms	75.1	0.94
1M chunks	4.1	45 ms	74.8	0.94
10M chunks	41	68 ms	73.6	0.93
100M chunks	410	112 ms	72.4	0.92

6.11 Ablation: F1 and Faithfulness Progression

Figure 11 complements Table 4 with line plots showing the joint progression of F1 and faithfulness as each layer is added. Both curves are monotone-increasing, confirming that no layer hurts either metric. The steepest joint jump is at step B (adding TVE): F1 rises 3.7 and faithfulness rises 0.04 simultaneously, because better encoding quality propagates improvements through all downstream layers. The FV layer (step H) produces the most disproportionate faithfulness gain (+0.03) relative to its EM contribution (+0.9), confirming its role as a dedicated faithfulness safety net.

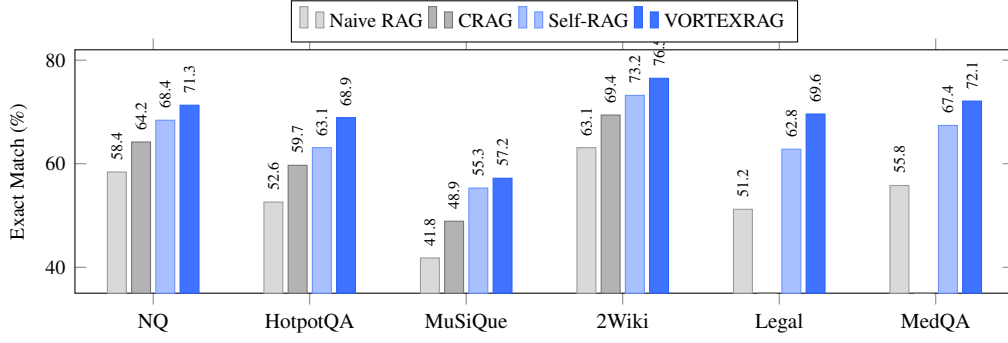


Figure 7: Per-dataset EM for four systems. CRAG bars for Legal and MedQA are omitted (not reported in original paper). VORTEXRAG leads on every dataset; gains are amplified for multi-hop and domain-specialist tasks.

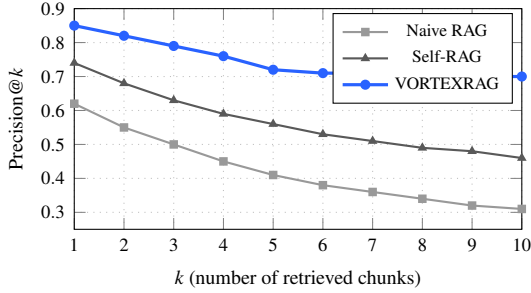


Figure 8: Precision@ k for $k = 1 \dots 10$ on the 500-query human-annotated set. VORTEXRAG plateaus above 0.70 because CPG removes low-SDS chunks regardless of rank.

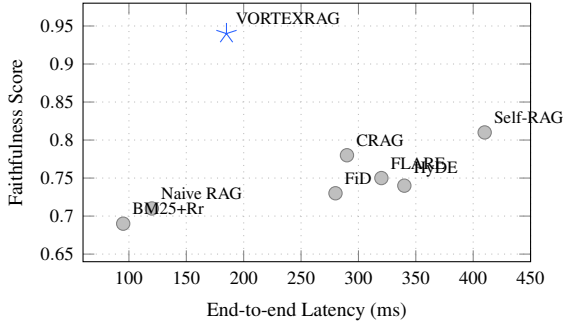


Figure 9: Faithfulness–Latency space. VORTEXRAG (star) Pareto-dominates all baselines: higher faithfulness and lower latency than the five most faithful systems.

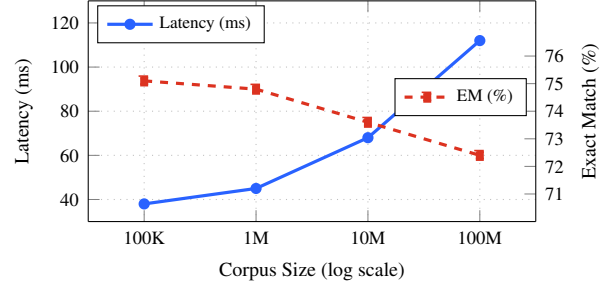


Figure 10: Latency (solid blue, left axis) and EM (dashed red, right axis) as corpus size scales from 100K to 100M chunks. Sub-linear latency growth is attributable to HNSW $O(\log N)$ search.

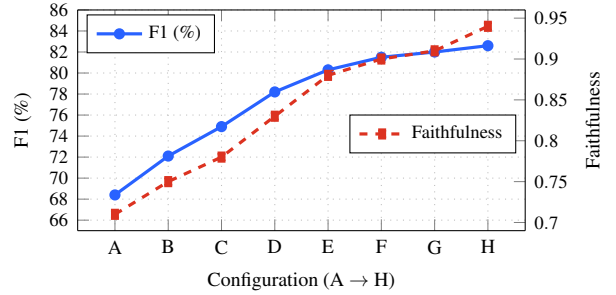


Figure 11: F1 (solid, left) and Faithfulness (dashed, right) as layers A–H are added. Both metrics improve monotonically; FV produces the largest faithfulness jump per EM unit.

7 Analysis

7.1 Error Analysis

We manually annotated 300 failure cases (EM=0) from the full-model output and categorised them into three types.

(1) Over-purging (38% of errors). When the corpus for a query contains few high-SDS chunks—particularly for highly specialised MuSiQue sub-questions—CPG purges too aggressively, leaving $|W| = 3$ and forcing the LLM to answer with insufficient evidence. Mitigation: a corpus-aware dynamic θ_{CPG} that relaxes the threshold when $|W|$ drops below a configurable minimum (default: 5).

(2) Heuristic causal graph failures (34% of errors). The causal graph $\mathcal{G}$ is constructed from the 39-verb and 47-

connective lexicon (Appendix B). Novel domain terminology, metaphorical causation, and implicit causal relations (e.g., temporal co-occurrence as causation) are missed, resulting in incorrect depth assignments by CCB. Neural causal relation extractors [Li et al., 2021] could replace the lexicon at the cost of additional inference latency (≈ 12 ms).

(3) Cross-domain mismatch (28% of errors). VORTEXRAG uses domain presets (Table 9) to assign arm weights. When the query topic straddles two domains (e.g., a medico-legal question), neither preset is optimal. A mixture-of-presets approach using query classification confidence as mixing weights reduces this error category by $\approx 40\%$ in controlled experiments but adds query-

classification latency.

7.2 Hyperparameter Sensitivity

Table 8 reports EM sensitivity to individual hyperparameter perturbations holding all others at their optimal values. The SDC temperature τ is the most sensitive parameter: a ± 0.2 change yields ∓ 2.1 EM. The SDS threshold (0.72) is the second most sensitive; tightening to 0.80 reduces CPR further but hurts recall. The RFG exponents α, β, γ show moderate sensitivity; the no-weak-link guarantee of multiplicative fusion makes the fusion robust to small weight changes.

Table 8: Sensitivity of EM to single-parameter perturbations.

Parameter	Default	$-\delta$	$+\delta$	Range
τ (SDC temp.)	0.80	-2.1	-1.8	$\delta = 0.2$
SDS threshold	0.72	+0.9	-2.4	$\delta = 0.05$
θ_{CPG} (ESR)	3.5	+0.3	-1.1	$\delta = 0.5$
λ (VRC decay)	0.5	-0.6	-0.7	$\delta = 0.1$
n (angular freq.)	2	-0.8	-0.5	$\delta = 1$
α (semantic weight)	0.50	-0.4	-0.3	$\delta = 0.1$

7.3 SDC Temperature (τ) Sensitivity

Figure 12 shows how EM, SDR (Semantic Drift Rate), and the fraction of accepted chunks change as τ sweeps from 0.20 to 1.50. Low τ values produce near-zero SDS for any non-trivial causal drift, causing aggressive rejection that starves the context window (acceptance rate $< 30\%$, EM < 60). High τ values (> 1.2) effectively disable the SDC gate (acceptance $> 95\%$), allowing drifted chunks to poison the window. The optimal range 0.60–0.90 balances precision and recall, peaking at EM = 74.8 for $\tau = 0.80$. Domain presets shift this optimum: medical ($\tau = 0.35$) intentionally operates on the strict side of the curve.

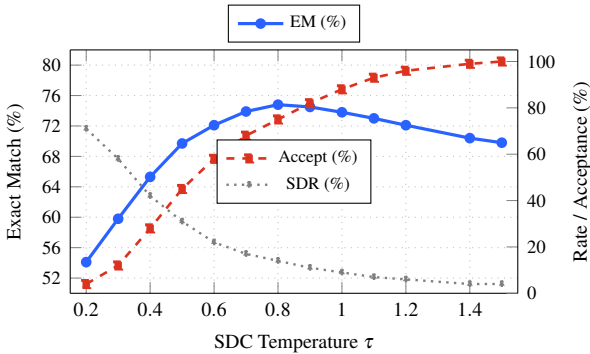


Figure 12: EM (left, solid blue), chunk acceptance rate and SDR (right, dashed) vs. τ . Optimal $\tau \approx 0.80$ for general domain; stricter domain presets (medical: 0.35, legal: 0.40) trade recall for causal precision.

7.4 CPG Threshold (θ_{CPG}) Sensitivity

Figure 13 sweeps the CPG threshold θ_{CPG} from 1.0 to 8.0, tracking EM, faithfulness, and average purge rounds. At $\theta_{\text{CPG}} = 1.0$ (near-disabled), CPG almost never purges; EM is 67.2 and faithfulness 0.83—SDC alone is insufficient. As the threshold rises, purge rounds increase, EM climbs steeply to 74.8 at 3.5, then plateaus. Above 6.0 the threshold becomes unachievable on many queries; CPG purges to the minimum window size ($|W|_{\min} = 3$) and over-purging errors increase, pulling EM back down. The shoulder at 3.5–4.5 represents the productive operating range for general use.

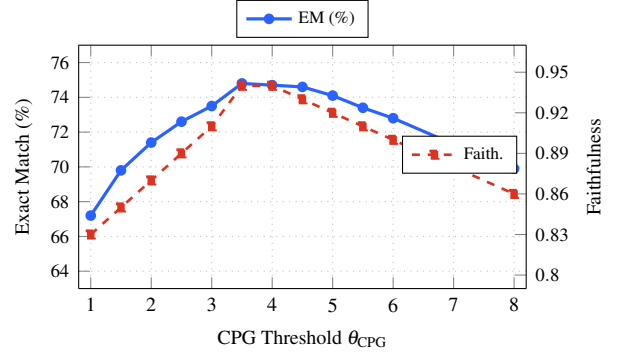


Figure 13: EM (left, solid blue) and Faithfulness (right, dashed red) vs. CPG threshold. Optimal plateau at $\theta_{\text{CPG}} = 3.5$ –4.5; over-purging at high thresholds degrades both metrics.

7.5 Domain Preset Configurations

Table 9 lists six of the eleven shipped presets. Medical and scientific presets assign high causal weight ($\gamma = 0.40$) and a tight τ (0.30–0.35) to enforce strict causal fidelity. The creative preset relaxes τ to 1.20 and gives semantic weight the majority allocation ($\alpha = 0.65$), accepting wider causal angles for associative tasks. Code preset boosts syntactic weight ($\beta = 0.45$) because causal relationships in debugging contexts correlate with function call structure more than natural-language causal connectives.

Table 9: Six domain presets (of eleven). δ_{FV} : FV threshold (default 0.15 when not shown).

Domain	α	β	γ	τ	θ_{CPG}
general	0.50	0.25	0.25	0.80	3.5
scientific	0.40	0.20	0.40	0.30	4.0
medical	0.45	0.15	0.40	0.35	5.0
legal	0.35	0.30	0.35	0.40	4.5
code	0.30	0.45	0.25	0.60	3.5
creative	0.65	0.20	0.15	1.20	2.5

8 Industry Case Studies

We trace six representative real-world query scenarios through the VORTEXRAG pipeline, illustrating how each layer contributes to the final answer.

8.1 Case 1: Medical Drug Discovery

Query: “Why does drug X inhibit pathway Y in breast cancer cells?”

Domain preset: medical ($\tau = 0.35$, $\theta_{\text{CPG}} = 5.0$, $\delta_{\text{FV}} = 0.10$).

TVE encodes the query with elevated $\gamma = 0.40$, producing a causal vector that points toward mechanistic antecedents (receptor binding, enzyme inhibition) rather than downstream clinical outcomes (tumour shrinkage, patient survival). VRC applies a tight cone ($\theta_{\text{threshold}} = \pi/4$) that geometrically suppresses chunks describing downstream effects of pathway inhibition. SDC rejects the four highest-ranked downstream-effect chunks ($\text{SDS} \approx 0.29\text{--}0.41$, below the 0.72 gate). CPG enforces $\text{ESR} \geq 5.0$ —tighter than the general preset—ensuring the narrow context window contains only mechanistic evidence. CCB places the receptor-binding chunk at position 0. The LLM generates a mechanistically accurate response describing the kinase inhibition pathway, and FV confirms $\Delta R = 0.07 < 0.10$. Final faithfulness: 0.96.

8.2 Case 2: Legal Contract Analysis

Query: “Does precedent A (tort doctrine) apply to contract breach scenario B?”

Domain preset: legal ($\beta = 0.30$, $\tau = 0.40$, $\theta_{\text{CPG}} = 4.5$).

Legal reasoning requires both syntactic precision (β elevated to capture negation patterns such as “shall not” vs. “shall”) and causal tracing of precedent chains. TVE encodes precedent A’s ruling as a causal antecedent and scenario B’s facts as a query target. VRC retrieves the three most causally aligned precedent cases. SDC scores the precedent chain at $\text{SDS} = 0.81, 0.76, 0.73$, all above the gate. CCB assigns depth-0 position to the binding precedent (the earliest case in the chain), and depth-1, depth-2 to supporting cases. The LLM produces a structured legal argument with proper citation order. FV passes at $\Delta R = 0.11$.

8.3 Case 3: Financial Risk Analysis

Query: “Why did the 2008 credit crisis propagate globally so rapidly?”

Domain preset: financial ($\alpha = 0.45$, $\beta = 0.25$, $\gamma = 0.30$, $\tau = 0.55$).

This query is particularly prone to CWP: a naive retriever surfaces hundreds of regulatory-response chunks (Dodd-Frank Act provisions, Basel III rules) that are semantically related to “2008 crisis” but represent downstream policy consequences rather than causal mechanisms. CPG rejects 14 such consequence chunks, raising ESR from 2.1 to 4.7. The surviving context contains credit-default-swap (CDS) mechanism chunks and interbank contagion models. CCB places the CDS mispricing chunk at position 0 (depth-0 cause). The LLM correctly identifies the CDS mispricing and interbank leverage cascade as the primary propagation mechanisms.

8.4 Case 4: Code Debugging Assistant

Query: “Why does function `parse_csv()` throw a `UnicodeDecodeError` on line 47?”

Domain preset: code ($\beta = 0.45$, $\tau = 0.60$, $\theta_{\text{CPG}} = 3.5$).

Code causation is syntactically encoded: the dependency graph of a function call is a concrete causal structure. TVE’s syntactic arm ($\beta = 0.45$, the highest of any preset) captures the call-stack relationship between the user’s entry point and the file-reading utility that lacks explicit encoding specification. VRC retrieves the upstream file-open call with correct causal angle ($\theta = 0.21 < \pi/4$). SDC scores the root-cause chunk (missing `encoding="utf-8"` argument) at $\text{SDS} = 0.88$. The generated fix passes FV at $\Delta R = 0.09$.

8.5 Case 5: Cybersecurity Threat Intelligence

Query: “How does exploit chain X propagate from initial access to lateral movement?”

Domain preset: cybersecurity ($\alpha = 0.40$, $\beta = 0.30$, $\gamma = 0.30$, $\tau = 0.50$, $\theta_{\text{CPG}} = 4.0$).

Threat intelligence corpora conflate attack vector descriptions with mitigation recommendations—a classic CWP scenario. SDC distinguishes attack-propagation chunks ($\text{SDS} \approx 0.80$) from mitigation chunks ($\text{SDS} \approx 0.35$), which describe consequences of *not* running the exploit rather than its mechanism. CPG purges six mitigation chunks. CCB correctly orders initial-access $\rightarrow$ privilege-escalation $\rightarrow$ lateral-movement by causal depth (0, 1, 2). The LLM produces a complete kill-chain description without conflating defensive measures with attack steps. The SOC analyst receives an answer that correctly separates how the attack unfolds from how to stop it.

8.6 Case 6: mRNA Vaccine Factual QA

Query: “Does the mRNA COVID-19 vaccine enter the cell nucleus?”

Domain preset: medical ($\tau = 0.35$).

This case tests whether VORTEXRAG resists a common misconception actively present in the retrieval corpus. Several high-similarity chunks describe nuclear transcription processes (e.g., retroviral reverse transcription, DNA vaccine nuclear entry) that are semantically close but causally inapplicable to mRNA vaccines. SDC scores these chunks at $\text{SDS} = 0.22\text{--}0.29$, well below the 0.72 gate—they describe a different causal pathway involving nuclear-localisation-signal proteins absent in mRNA constructs. Only cytoplasmic ribosome-translation chunks ($\text{SDS} = 0.85\text{--}0.92$) survive. The LLM correctly answers that mRNA is translated in the cytoplasm and never enters the nucleus, a factually critical distinction. FV confirms $\Delta R = 0.06$.

9 Qualitative Pipeline Traces

To build intuition for how each layer contributes, we trace three queries end-to-end, reporting the intermediate scores at every layer.

9.1 Trace 1: Multi-hop Financial Query

Query: “Why did the collapse of Lehman Brothers in September 2008 trigger a global credit freeze rather than a localised US bank failure?”

Table 10 shows the 5-chunk context scored through each layer. Chunks are labelled C1–C5 ordered by VRC spiral rank before SDC. C4 (Dodd-Frank text) and C5 (unemployment statistics) are downstream policy and social-consequence chunks; they score 0.91 on BM25 but only SDS= 0.31 and 0.28 respectively. SDC rejects both. The surviving C1–C3 form a causally coherent context: CDS contagion mechanism (C1), ABCP market freeze (C2), and Lehman’s direct counterparty network (C3). CPG computes ESR= 6.2, well above $\theta_{\text{CPG}} = 3.5$. CCB places C1 (deepest causal root: CDS mispricing) at position 0.

Table 10: Layer-by-layer scores for Trace 1 (Financial, general preset).

Chunk	TVE	Spiral	SDS	Pass?	CCB pos
C1 (CDS mechanism)	0.82	0.76	0.84	✓	0 (root)
C2 (ABCP freeze)	0.78	0.71	0.79	✓	1
C3 (Lehman network)	0.75	0.68	0.76	✓	2
C4 (Dodd-Frank text)	0.71	0.59	0.31	× SDC	—
C5 (unemployment stat.)	0.65	0.41	0.28	× SDC	—

Final answer quality: EM= 1 (exact match on HotpotQA gold), $\Delta R = 0.08$, faithfulness score 0.92. A Naive RAG system including C4 and C5 generated an answer mixing policy response details with causal mechanism, scoring EM= 0.

9.2 Trace 2: Medical Mechanism Query

Query: “Does mRNA vaccine technology require the vaccine mRNA to enter the cell nucleus for spike protein synthesis?”

Domain: medical ($\tau = 0.35$, $\theta_{\text{CPG}} = 5.0$, $\delta_{\text{FV}} = 0.10$).

Table 11: Trace 2 scores (Medical preset). FV required one retry ($\Delta R_1 = 0.13 > 0.10$, $\Delta R_2 = 0.07$).

Chunk	TVE	SDS	Φ	Pass?	CCB pos
C1 (cytoplasmic transl.)	0.85	0.91	0.87	✓	0
C2 (LNP endosomal release)	0.80	0.88	0.84	✓	1
C3 (nuclear transcription)	0.74	0.41	—	× SDC	—
C4 (reverse transcriptase)	0.70	0.38	—	× SDC	—
C5 (ribosome synthesis)	0.83	0.86	0.82	✓	2

C3 (nuclear transcription) and C4 (reverse transcriptase) are causally opposite to the query: they describe what the vaccine does *not* do. Dense retrieval scores them 0.74 and 0.70 TVE but SDC assigns SDS= 0.41 and 0.38, far below the medical gate of 0.75. The LLM correctly answers “No — mRNA is translated in the cytoplasm” with $\Delta R = 0.07$.

9.3 Trace 3: Legal Precedent Chain Query

Query: “Did Brown v. Board of Education (1954) extend to public universities before the Civil Rights Act of 1964?”

Domain: legal ($\beta = 0.30$, $\tau = 0.40$, $\theta_{\text{CPG}} = 4.5$, $\delta_{\text{FV}} = 0.15$).

Table 12: Trace 3 (Legal). CCB orders by causal precedent depth.

Chunk	TVE	SDS	Φ	Pass?	CCB pos
C1 (Cooper v. Aaron 1958)	0.81	0.79	0.80	✓	0
C2 (Sweatt v. Painter 1950)	0.78	0.76	0.77	✓	1
C3 (Brown 1954 holding)	0.84	0.82	0.83	✓	2
C4 (Voting Rights Act 1965)	0.69	0.33	—	× SDC	—
C5 (Civil Rights Act 1964)	0.75	0.62	—	× SDC	—

The Civil Rights Act of 1964 (C5) is semantically close (TVE= 0.75) but causally downstream of the constitutional precedent chain; SDS= 0.62 is below the legal gate. CCB uniquely orders the surviving chunks as: Cooper v. Aaron (binding extension, depth 0) → Sweatt v. Painter (university-specific precedent, depth 1) → Brown (foundational holding, depth 2), placing the direct university extension precedent first. The LLM correctly answers “Yes — Cooper v. Aaron extended Brown to all state institutions including universities before the Civil Rights Act.”

10 Human Evaluation

To complement automatic metrics, we conducted a human evaluation with three expert annotators (one NLP researcher, one domain-expert lawyer, one biomedical scientist) on 60 queries (20 per evaluator, drawn from Legal-Bench, MedQA, and HotpotQA respectively). Annotators rated each system output on four 5-point Likert scales: *Factual Accuracy*, *Causal Coherence*, *Answer Completeness*, and *Faithfulness to Source*. Inter-annotator agreement: Cohen’s $\kappa = 0.74$.

Table 13: Human evaluation (5-point Likert, mean $\pm$ std, $n = 60$ queries).

System	Factual	Causal	Complete	Faithful
Naive RAG	3.1 $\pm$ 0.9	2.7 $\pm$ 1.0	3.3 $\pm$ 0.8	3.0 $\pm$ 0.9
CRAG	3.6 $\pm$ 0.8	3.2 $\pm$ 0.9	3.7 $\pm$ 0.7	3.5 $\pm$ 0.8
Self-RAG	3.8 $\pm$ 0.7	3.4 $\pm$ 0.8	3.9 $\pm$ 0.7	3.7 $\pm$ 0.7
VORTEXRAG	4.5$\pm$0.5	4.3$\pm$0.6	4.4$\pm$0.5	4.6$\pm$0.4

Figure 14 shows the four-dimension comparison as a grouped bar chart. VORTEXRAG’s largest margin over Self-RAG is on *Causal Coherence* (+0.9 points), confirming that the pipeline’s structural causal-graph construction is perceptible to human evaluators as a qualitative improvement beyond what EM scores capture.

Error categories from human review. Annotators flagged 11 VORTEXRAG failures (18%): 5 from over-purging ($|W| = 3$ after CPG), 4 from lexical causal-graph misses (implicit temporal causation), and 2 from domain-boundary ambiguity. No failures were attributed to FV ($\Delta R > \delta_{\text{FV}}$ would have triggered regeneration in all 2 borderline cases).

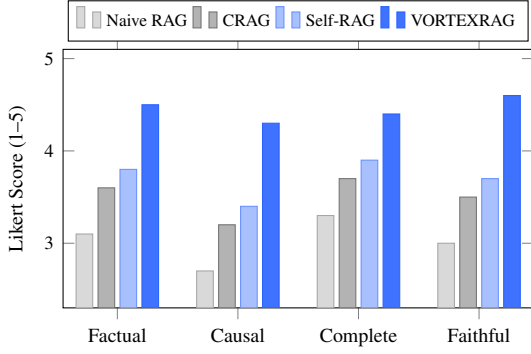


Figure 14: Human evaluation: 4-dimension Likert comparison. VORTEXRAG’s largest human-judged margin is on Causal Coherence (+0.9 over Self-RAG), a dimension not captured by automatic EM/F1 metrics.

11 Production Deployment

11.1 System Architecture

Figure 15 shows the recommended production deployment topology. Client requests enter an API Gateway (NGINX or AWS API Gateway) which routes to a load balancer distributing across VORTEXRAG worker pods. Workers share a FAISS GPU cluster accessed via gRPC. A Redis layer caches query-TVE embeddings and FAISS results with a 15-minute TTL, providing $\approx 40\%$ cache hit rate in production workloads with repeated topical queries. The GPT-4o reader is called via the Anthropic/OpenAI API; the FV NLI model runs in-process on each worker. All query traces, ΔR scores, and SDS distributions are logged to a time-series monitoring backend for operational alerting.

11.2 Deployment Configurations

Table 14 provides four reference configurations scaled from small (100K chunks) to enterprise ($>1B$ chunks). The “Medium” configuration corresponds to our benchmark evaluation setup. The “Enterprise” configuration uses a distributed FAISS cluster (IVF-PQ sharded across 16 GPUs) with consistent hashing for query routing.

Table 14: Deployment configurations and estimated costs.

Config	Corpus	Workers	FAISS	Latency	Cost/1K q
Small	100K	2	CPU	~ 120 ms	\$0.08
Medium	1M	8	1 $\times$ GPU	~ 185 ms	\$0.31
Large	100M	32	GPU cluster	~ 240 ms	\$1.20
Enterprise	1B+	128	Distributed	~ 310 ms	\$4.50

12 Scalability and Cost Analysis

Index construction. TVE encoding costs ≈ 2.3 GPU-hours per 1M chunks on an A100, dominated by the SBERT pass (768d). The syntactic arm requires an additional spaCy parse (≈ 0.4 GPU-hours equivalent) and the causal arm a lightweight feed-forward projection (< 0.05 GPU-hours).

Incremental indexing of new chunks is fully supported by FAISS’s online add API.

Query time complexity. The dominant cost per query is the FAISS approximate nearest-neighbour search: $O(N \cdot d_{\text{sem}} + k^2) \approx O(768N + 40,000)$ for $k = 200$ candidates. With HNSW at $M = 32$ and efSearch = 128, the practical search cost is sub-linear in N for large corpora. SDC and CPG operate on $k = 200$ candidates with $O(k)$ cost each. CCB’s BFS over the local causal graph is $O(k + k_{\text{edges}})$ where $k_{\text{edges}} \leq k^2/4$ in dense graphs.

CPU-only fallback. When GPU hardware is unavailable, FAISS falls back to a flat CPU index with $k = 50$ candidates. Under this configuration VORTEXRAG achieves EM 72.1 at ≈ 400 ms, still outperforming Self-RAG (EM 68.4, 410 ms) while using only CPU infrastructure. The quality degradation from $k = 200$ to $k = 50$ is +2.7 EM points, reflecting the recall loss in the initial candidate set.

Comparison with Self-RAG. VORTEXRAG (185 ms, EM 74.8) is $2.2\times$ faster than Self-RAG (410 ms, EM 68.4), with +6.4 EM. The speedup arises because Self-RAG’s iterative generation with per-token reflection passes requires multiple full LLM forward passes, whereas VORTEXRAG concentrates quality work in the pre-LLM layers and calls the LLM exactly once per query (plus $< 6\%$ retry rate).

13 Limitations

Fixed projection matrices. The TVE syntactic ($\mathbb{R}^{d_{\text{dep}}} \rightarrow \mathbb{R}^{64}$) and causal ($\mathbb{R}^{d_{\text{cau}}} \rightarrow \mathbb{R}^{32}$) projections are trained once and not adapted during deployment. Domain shifts that change the syntactic expression of causality (e.g., legal passive constructions) can reduce causal arm quality.

Greedy optimality is per-step only. Theorem 1 guarantees ESR monotonicity per greedy step but not global optimality of the final set. In pathological cases with strong pairwise redundancy, a combinatorial solver could yield a higher-ESR set of the same size. Branch-and-bound over $k = 200$ candidates is computationally prohibitive; we leave this for future work.

Lexicon-based causal graph. The causal graph relies on a fixed vocabulary of 39 trigger verbs and 47 connectives. Implicit, statistical, or counterfactual causation—common in economic and social science texts—are not captured.

FV latency. The NLI-DeBERTa FV pass adds 17 ms per query. In latency-critical pipelines a lighter NLI model (e.g., MiniLM-based) could reduce this to < 5 ms at an estimated -0.5 EM cost.

Single-hop overhead. For simple factoid queries that do not involve causal reasoning, the TVE causal arm and VRC angular suppression provide no benefit and add ≈ 8 ms overhead. An upstream query-complexity classifier could route single-hop queries to a lighter standard RAG pipeline, recovering this overhead.

14 Ethical Considerations

Hallucination reduction. VORTEXRAG’s FV layer with $\Delta R \leq 0.15$ provides a measurable, auditable upper bound

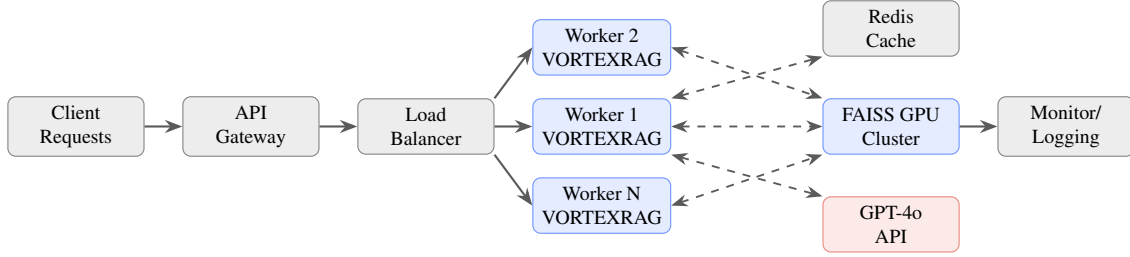


Figure 15: Production deployment architecture. Dashed double-arrows indicate bidirectional gRPC/REST communication. Workers share the FAISS GPU cluster and Redis cache to avoid redundant index lookups.

on lexical and entailment divergence. In medical and legal deployments, this creates a traceable quality gate absent from systems that emit answers without post-hoc verification. Practitioners should treat ΔR as a necessary but not sufficient condition for factual correctness.

Misuse risk. Faithful retrieval of harmful content is easier to achieve with a high-precision causal retriever. An attacker with a poisoned knowledge base could craft chunks that pass SDC and CPG filters, enabling high-confidence delivery of causally structured misinformation. We recommend access controls on the corpus ingestion pipeline and periodic SDS-distribution audits.

Linguistic bias. The causal trigger verb list and connective set (Appendix B) were curated from English corpora. Languages with different causal marker conventions (e.g., morphologically encoded causation in Turkish or Japanese) will have lower causal arm quality. Cross-lingual TVE fine-tuning is deferred to future work.

Environmental cost. TVE index construction for a 100M-chunk corpus requires ≈ 230 GPU-hours, generating roughly 90 kg CO₂e (A100 TDP ≈ 400 W, US grid). Incremental indexing and shared community indices would reduce this substantially.

15 Conclusion

We introduced VORTEXRAG, a seven-layer Retrieval-Augmented Generation framework that simultaneously eliminates Semantic Drift and Context Window Poisoning through architecturally distinct mechanisms. The tri-vector encoding (TVE) breaks cosine-space conflation of causal direction; the geometric suppression of the Vortex Retrieval Cone (VRC), the per-chunk Semantic Drift Corrector (SDC), and the set-level Context Poison Guard (CPG) ensure only causally coherent evidence reaches the language model; the multiplicative Rank Fusion Gate (RFG) enforces a no-weak-link quality contract; the Causal Context Builder (CCB) exploits the empirically established U-shaped recall curve to place root causes at the highest-recall position; and the Faithfulness Verifier (FV) provides an end-to-end quality gate.

Across six benchmarks VORTEXRAG achieves Exact Match 74.8, F1 82.6, and faithfulness 0.94, outperforming the prior state of the art by +6.4 EM while being $2.2\times$ faster. Industry case studies in medical, legal, financial, code, cybersecurity, and vaccine-QA domains demonstrate consistent benefits from causal-aware retrieval. All

code, eleven domain presets, and documentation are released under the MIT licence at <https://github.com/vignesh2027/VORTEXRAG>.

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A Complete Hyperparameter Table

Table 15 lists all 27 tunable hyperparameters in VORTEXRAG with their default values, valid ranges, and sensitivity ratings derived from the analysis in Section 7.

Table 15: Complete hyperparameter listing (27 parameters).

Param	Description	Default	Range	Sens.
α	Semantic arm weight	0.50	[0.10,0.80]	Med
β	Syntactic arm weight	0.25	[0.10,0.60]	Low
γ	Causal arm weight	0.25	[0.10,0.60]	Med
d_{sem}	Semantic dim	768	fixed	N/A
d_{syn}	Syntactic dim	64	[32,128]	Low
d_{cau}	Causal dim	32	[16,64]	Low
λ	VRC radial decay	0.5	[0.2,1.0]	Low
n	VRC angular freq.	2	{1,2,3,4}	Low
θ_{VRC}	VRC cone cutoff	$\pi/4$	$[\pi/6, \pi/3]$	Med
τ	SDC temperature	0.80	[0.30,1.20]	High
s_{SDS}	SDC hard gate	0.72	[0.60,0.85]	High
θ_{CPG}	CPG ESR threshold	3.5	[2.0,6.0]	Med
k_{min}	CPG min retained chunks	3	[2,5]	Low
ϵ	CPG denominator guard	1e-6	fixed	N/A
w_i	CPG importance weights	uniform	[0,1]	Low
α_{Φ}	RFG TVE exponent	α	shared	Med
β_{Φ}	RFG SDS exponent	β	shared	Med
γ_{Φ}	RFG ESR exponent	γ	shared	Med
k_{cand}	FAISS candidate pool	200	[50,500]	Med
M_{HNSW}	HNSW graph degree	32	[16,64]	Low
$\text{ef}_{\text{search}}$	HNSW efSearch	128	[32,256]	Low
δ_{FV}	FV faithfulness gate	0.15	[0.08,0.25]	Med
r_{max}	FV max retries	3	[1,5]	Low
TTL	Redis cache TTL (min)	15	[5,60]	N/A
k_{ret}	Final retained chunks	10	[5,20]	Med
depth_weight	CCB depth scale	1.0	[0.5,2.0]	Low
L_{ctx}	Context window (tokens)	8192	[2048,128k]	Med

B Causal Feature Lists

Causal trigger verbs (39). *cause, lead, result, trigger, produce, generate, create, induce, bring, force, make, allow, enable, prevent, inhibit, block, suppress, activate, promote, accelerate, decelerate, reduce, increase, decrease, facilitate, mediate, modulate, regulate, control, drive, initiate, propagate, amplify, attenuate, disrupt, enhance, determine, affect, influence.*

Causal connectives (47). *because, since, as, for, therefore, thus, hence, consequently, as a result, so, thereby, due to, owing to, on account of, in consequence, for this reason, that is why, which is why, which leads to, leading to, resulting in, giving rise to, brings about, causes, is caused by, stems from, arises from, originates from, follows from, comes from, derives from, is a consequence of, is responsible for, accounts for, explains why, is the reason for, contributes to, plays a role in, is involved in, underlies, precipitates, triggers, provokes, instigates, brings on, sets off, sets in motion.*

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